Discriminative Reward Policy Optimization for Relational Reasoning in Language-Driven Robotic Grasping

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***Abstract*—Language-driven robotic grasping aims to turn free-form language instructions and visual perception into executable 6-DoF actions. Recent works have integrated Large Language Models (LLMs) or Vision-Language Models (VLMs) into grasping frameworks, achieving promising results in standard tabletop scenarios. However, their accuracy drops dramatically in cluttered environments with strong intra-class interference. This limitation could be attributed to a fundamental objective mismatch: supervision based on a single annotation neither enforces within-scene ranking among near-duplicate instances nor incorporates non-differentiable executability feedback. The model thus tends to memorize training patterns rather than perform explicit relational reasoning, restricting its accuracy and generalization in complex, cluttered scenarios. To tackle these issues, we reformulate candidate selection as a top-k decision process and introduce Discriminative Reward Policy Optimization (DRPO), a reinforcement learning paradigm that optimizes rule-verified rewards capturing spatial localization, semantic alignment, intra-class disambiguation, and grasp executability. The policy is guided by a discriminative reward that improves to identify and select the correct object among visually similar candidates. Consequently, our method improves the success rate in distinguishing visually similar instances from the same category, achieving a 10% gain on the ambiguous split of FreeGraspData. This demonstrates that a dedicated reward, combined with DRPO, enhances the reasoning capabilities of language-driven grasping and restores practical accuracy. The code and trained models will be made publicly available.**

***Index Terms*—Robotic vision, grasp detection, language-guided detection, reinforcement fine-tuning.**

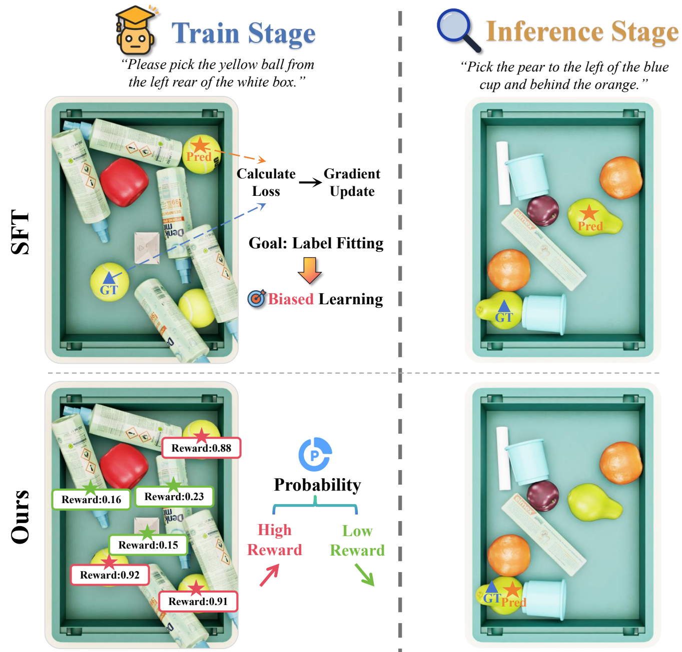

Fig. 1. Comparison of Supervised Fine-tuning (SFT) and our method across both training and inference stages in complex scenes that contains multiple instances of same-category. **During training** under single-label supervision, the model is mainly optimized to match the annotated target, with limited guidance on how to distinguish it from other visually or semantically similar candidates in the same scene. This may bias the model toward dataset-specific appearance or location cues rather than robust fine-grained discrimination; in contrast, our method scores region–pose candidates with task-verifiable rewards, shifting probability toward high-reward instances. **At the inference phase,** our method achieves higher success rate in this scenario.

I. Introduction

TRADITIONAL robotic grasping [1]–[4] primarily relies on visual geometry and affordance cues to predict feasible grasp poses. While this enables reliable manipulation in structured environments, it remains limited to geometry or category-based targets. With the rapid advancement of natural language understanding and vision–language grounding, language-driven robotic grasping has emerged as a semantic-level evolution of this line of research, allowing robots to interpret natural-language instructions for goal-directed manipulation. In recent years, the language-driven robotic grasping task has evolved from simple scenes containing a single object [5], [6] to cluttered environments with multiple objects [7]. Early studies [5], [6], [8] primarily addressed language instructions referring to the target object by category, category+attributes (e.g., color, size), or category+attributes+explicit spatial locations [9]–[11]. Recent foundation-model-based methods [12], [13] have further explored open-vocabulary instance-level grasping. For example, RoG-SAM [12] adapts SAM [14] to localize user-specified object instances and predict grasp poses from language prompts, showing the potential of promptable visual models for interactive grasping. More recently, free-form language instructions [15], which are naturally used by humans in daily life, have been introduced; however, such descriptions are often *ambiguous*, especially when multiple instances of same-category or visually similar objects are present [16], as illustrated in Fig. 1.

To date, *the challenge of coexisting near-duplicate instances in complex real-world scenes remain under-explored.* Some prior works [17] leverage vision-language models (VLMs) such as GPT-4o [18] to identify the correct target object; however, these models frequently suffer from hallucinated or inconsistent spatial relationships during reasoning. Free-Grasp [15] addresses this issue by using a structured prompt

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that specifies key elements in the instruction, including the task setup, objective, and reasoning logic. Nonetheless, its effectiveness may be limited in scenarios that require fine-grained visual grounding, as structured prompting does not directly strengthen the model's grounding ability. In practice, existing VLM-based methods [9], [15], [19] rely heavily on supervised fine-tuning (SFT) to acquire implicit relational reasoning abilities, as depicted in Fig. 1. This paradigm typically optimizes the likelihood of selecting the correct prediction from a set of candidates. However, single-annotation supervision may provide limited guidance for ranking visually similar instances within the same scene. As a result, this process may encourage the model to fit dataset-specific patterns rather than explicitly comparing visually similar candidates, which can limit its accuracy and generalization in open-world scenarios with multiple objects from the same category. This naturally raises the following question: How can we leverage existing data and learning paradigms to effectively enhance robotic grasping?

To address this question, we reformulate the problem as a top-k selection task and leverage reinforcement learning with **rule-defined, verifiable rewards** to reason the final grasp, rather than simply maximizing the probability of selecting among candidate predictions.

This idea is instantiated in our proposed **Discriminative Reward Policy Optimization (DRPO)**, which aims to reliably enhance relational reasoning in language-driven robotic grasping. DRPO uses four task-specific rewards, including **spatial localization**, **semantic alignment**, **intra-class disambiguation**, and **grasp executability**. Each reward is defined by a verifiable criterion. For each instruction, the policy samples a small group of candidate grasps and evaluates them using task-verifiable rewards. The reward scores are normalized within the group to compute advantages, and the policy is updated with a clipped likelihood-ratio objective to a reference model. In this way, the optimization is directly aligned with the desired grasp behaviors, enabling the model to learn from ranking-based, non-differentiable, and task-verifiable feedback, such as grasp executability and spatial consistency, which are difficult to capture with conventional SFT. To systematically evaluate our proposed method, we conduct extensive experiments on both relational-reasoning benchmarks and real-world scenarios. Our method achieves state-of-the-art (SOTA) performance in Reasoning Success Rate (RSR) on the public FreeGraspData dataset [15], especially in ambiguous split, retaining 10% gains. Furthermore, we perform real-robot experiments in which our method consistently successfully grasps the target object among multiple same-category instances specified by language instructions, demonstrating a very high success rate in practical execution.

In general, our contributions are summarized as follows:

1) We introduce a reinforcement learning-based approach, *Discriminative Reward Policy Optimization*, to enhance relational reasoning in language-driven robotic grasping.
2) We design four rule-defined and task-verifiable rewards, i.e., spatial localization, semantic alignment, intra-class disambiguation, and grasp executability, to address ambiguities in complex, multi-object scenes and directly align optimization with grasp success signals.
3) Through extensive evaluations on relational reasoning benchmarks and real-world robot experiments, our method achieves SOTA performance in reasoning accuracy and consistently demonstrates robust grasp execution under language-guided instructions.

II. Related Work

A. Language-driven robotic grasping

Recent advances [20] have focused on grasp pose prediction from language inputs. Diffusion-based models [7], [21] have shown potential by learning probabilistic processes to generate grasp pose. Leveraging the benefits of object-centric representations for enhanced scene understanding and grasp planning, some methods jointly learn segmentation and grasp detection. For example, Tziafas et al. [22] show that jointly training segmentation and grasp estimation in an end-to-end manner improves both segmentation quality and grasp success rates. RelationGrasp [9] uses object prompts to unify grasp detection and relation inference in open-vocabulary scenes. Yang et al. [23] proposed an attribute-based robotic grasping method that combines workspace images with query text to predict grasp poses. Mei et al. [12] proposed RoG-SAM, a language-driven framework that equips SAM with lightweight adapters and a grounding detector for instance-level grasp pose prediction. In contrast to these approaches, our work introduces a novel end-to-end trainable network that incorporates candidate-level region–pose modeling to further boost grasping performance.

B. VLM-driven robotic grasping

Motivated by the reasoning capabilities of LLMs [24]–[27], recent work has adapted these abilities to robotics by combining language models with visual encoders, yielding VLMs [28] for language-conditioned grasping. Language reasoning alone is insufficient for real-world manipulation, as robots must convert instructions into spatially executable actions. To this end, prior studies [10], [29] ground natural-language instructions into spatial affordances and explicit spatial representations, linking object-interaction cues to graspable regions. For example, RoboPoint [10] and RoboRefer [11] predict affordance points from language instructions, and RT-grasp [6] introduces an intermediate reasoning phase to refine grasp pose generation, though it is often impractical for real-time execution. More recent methods [15], [17] leverage GPT-4o [18] for language-based reasoning, with FreeGrasp [15] employing iterative Chain-of-Thought (CoT) occlusion reasoning. However, these approaches remain limited in grounding multiple same-category objects in complex, cluttered scenes.

C. Reinforcement Fine-tuning

Reinforcement learning has been explored as a complement to imitation for visuomotor grasping by directly optimizing task rewards. VPG [30] provides evidence that deep Q-learning with self-supervised signals improves performance under occlusion and partial observability in the reported setting. When

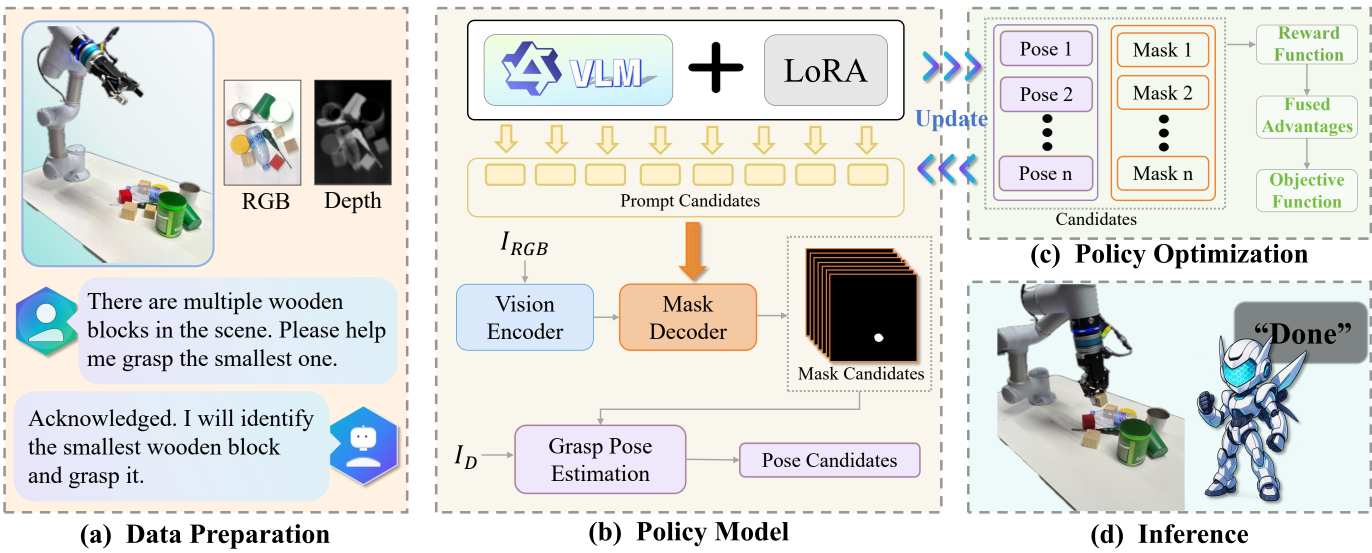

Fig. 2. **The overall framework of the proposal. (a)** For each language-driven grasp, a RGB-D scene O is paired with an instruction g, hence (g, O) is fed to the model. **(b)** During training, a LoRA-adapted VLM proposes instruction-aligned prompt candidates; a vision encoder and mask decoder yield instance masks; from the depth map I_D, local point clouds are built and a grasp estimator predicts pose candidates. **(c)** Mask–pose pairs form the candidate set for DRPO: a reward R scores each pair, scores are group-standardized into fused advantages, and the policy is updated by the DRPO objective. **(d)** At inference, the policy selects the target 6-DoF pose for execution.

grasp policies operate in high-DoF continuous action spaces, extending value-based methods is less straightforward due to action discretization error and instability in Q-targets. In this regime, PaG [31] adopts a policy-gradient formulation with PPO [32], where the clipped surrogate objective yields more robust updates for continuous control. Beyond PPO for continuous manipulation, reinforcement fine-tuning adapts policy-gradient objectives to preference-based feedback, this line of work replaces hand-designed dense rewards with preference signals, while retaining policy-gradient optimization. GRPO, proposed by DeepSeek [33] and used in R1 [34], aligns pretrained models via groupwise relative preferences with minimal online interaction in practice at scale.

III. Methodology

A. Overview

To enable language-driven grasping, we propose a new grasping scheme that predicts 6-DoF poses from RGB-D images [35]. In scenarios involving multiple similar objects, most current practices of target grasping struggle to maintain high precision. Accordingly, we propose a scheme that reliably resolves such ambiguities and remains robust in cluttered scenes, ensuring effective operations of grasping [36]. As illustrated in Fig. 2, we cast the pose-prediction module as a language-grounded policy model that integrates both a VLM and a grasp estimation component. Given a language instruction g and an RGB–D observation O, the policy first employs the VLM to generate a semantically enriched visual prompt, represented as a 2D bounding box in the RGB image predicted from (g, O), providing coarse spatial grounding of the referred object. Conditioned on this prompt, the segmentation stage first generates fine-grained object masks. The grasp estimation component then combines each mask with depth information to infer the corresponding grasp pose.Both predicted masks and grasp poses are subsequently used for model training.

The concerned policy is trained with the DRPO methodology, as shown in Fig. 3. Given each instruction, we build a candidate group from the policy model and score it with a reward function that jointly considers language grounding, in-scene disambiguation, and pose geometry. We optimize a clipped likelihood-ratio objective with group-normalized advantages and a Kullback-Leibler (KL) [37] term to a supervised reference. During the inference stage, the policy is executed in a single shot that directly outputs a 6-DoF grasp pose.

B. Language-Grounded Policy Architecture

1) VLM-based Semantic Localization: The policy incorporates instruction-aligned semantic cues from a Qwen-based VLM model [25], using both the instruction and RGB–D observation as inputs. The LoRA adapter [38] is introduced in the language and cross-modal layers for fine-tuning. The adapted VLM serves as a trainable semantic module for grasping, helping the model handle cluttered scenes with visually similar objects.

2) Fine-Grained Object Segmentation: The language-grounded prompts produced by the VLM are routed to a Fine-Grained Object Segmentation [14], achieving a set of masks that are consistent with the given instruction. Owing to its promptable design, inherited from SAM, the segmenter can accept prompts and generate plausible masks even for ambiguous inputs. This capability makes it a natural and composable component within our pipeline. A frozen vision encoder first extracts image features. A prompt-conditioned mask decoder then uses these features together with the prompts generated by the VLM to predict one fine-grained mask for each prompt.

3) Pose Estimation for Grasp: For each retained mask, we back-project its pixels using the depth channel and calibrated camera intrinsics to obtain an instance-specific point cloud. The mask-cropped point cloud is fed to Grasp Pose Estimation

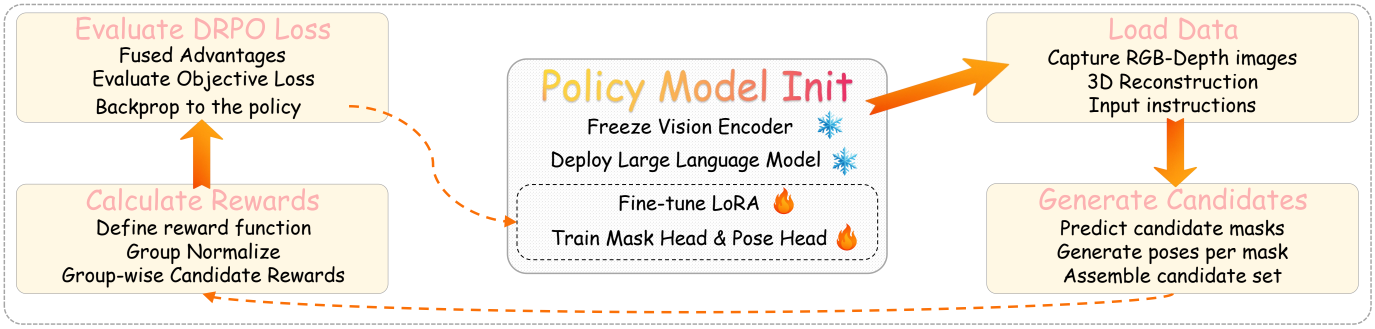

Fig. 3. Training procedure of DRPO. Each iteration sequentially initializes the policy, loads data, generates mask–pose candidates, computes group-normalized rewards, and evaluates the DRPO objective loss. Ultimately, the loss is backpropagated to update the policy.

stage [2] to regress grasp poses: the network produces 6-DoF hypotheses $p = (\mathbf{t}, \mathbf{R}, w)$, where $\mathbf{t}$ and $\mathbf{R}$ denote translation and rotation, and w is the gripper width. These predictions constitute the Pose Candidates used for DRPO method.

C. Discriminative Optimization of Reward Policy

We optimize a language–grounded grasping policy π_θ over a discrete set of region–pose candidates. For a query $q = (O, g)$, the generator produces a group of N candidates. Each one comprises a segmentation mask m and a 6-DoF pose p, denoted as $o_i = (m_i, p_i)$. The policy model assigns probabilities $\pi_\theta(o_i \mid q)$ to this group. The proposed Discriminative Reward Policy Optimization (DRPO) learns an candidate ranking by standardizing rewards within the scene and updating the categorical policy so that higher-reward candidates receive higher selection probability, with a clipped ratio and a KL-to-reference constraining update.

1) Group Rewards and Standardization: Each candidate is evaluated by a terminal reward as follows:

$$r_i \;=\; R\big(O, g, m_i, p_i\big). \tag{1}$$

We design the reward function as a composite of four verifiable components tailored to our task. These components encompass: spatial location, semantic alignment, intra-class disambiguation, and grasp executability (Sec. III-C4).

In scenarios with significant intra-class interference, the reward function R is used to score each candidate. We further apply group-wise standardization to the rewards; this eliminates scene-level scale/shift variance, thereby favoring higher-reward candidates within the scene.

The group-wise normalization yields an advantage A_i that preserves the in-scene ordering of candidates:

$$A_i \;=\; \frac{r_i - \mathrm{mean}\{r_1, \ldots, r_N\}}{\mathrm{std}\{r_1, \ldots, r_N\} + \varepsilon}. \tag{2}$$

2) Likelihood Ratios and Clipped Surrogate: The behavior policy $\pi_{\theta_{\text{old}}}$ samples the group. The sequence–level likelihood ratio and its clipped counterpart are represented as:

$$\begin{aligned} s_1 \;&=\; \frac{\pi_\theta(o_i \mid q)}{\pi_{\theta_{\text{old}}}(o_i \mid q)}, \\ s_2 \;&=\; \mathrm{clip}\left(\frac{\pi_\theta(o_i \mid q)}{\pi_{\theta_{\text{old}}}(o_i \mid q)},\; 1-\epsilon,\; 1+\epsilon\right), \end{aligned} \tag{3}$$

where ϵ is a hyper-parameter [32]. These ratios are computed with the same tokenization and factorization as at inference so that training and deployment operate on identical probability models. The per–candidate surrogate combines the advantage with the clipped update, such as:

$$\mathcal{Q}(i) = \min\big[s_1 A_i,\; s_2 A_i\big], \tag{4}$$

which bounds the effective step size in probability space and reduces variance. Intuitively, $\mathcal{Q}(i)$ increases the likelihood of candidates with positive A_i while preventing unstable updates when the raw ratio s_1 is large. For negative A_i, both terms reduce the selection probability, with the same clipping constraint to keep the update stable.

3) Reference Regularization and Final Objective: To preserve language grounding and geometry priors acquired by supervised learning, we then impose a Kullback–Leibler regularizer [37]:

$$\mathcal{R}(\theta) = \beta\, \mathbb{D}_{\mathrm{KL}}\big[\pi_\theta \,\|\, \pi_{\text{ref}}\big]. \tag{5}$$

The coefficient β controls the trade–off between exploiting group–relative returns and staying close to the supervised reference π_{ref}. A larger β yields conservative updates that mitigate distribution drift; a smaller β allows stronger adaptation to the task return. The DRPO objective averages the surrogate over the group and subtracts the KL penalty, i.e.,

$$\mathcal{J}(\theta) = \mathbb{E}_{\,o_{1:N} \sim \pi_{\theta_{\text{old}}}(\cdot \mid q)} \left[\frac{1}{N} \sum_{i=1}^{N} \Big(\mathcal{Q}(i) - \mathcal{R}(\theta) \Big) \right]. \tag{6}$$

Optimizing $\mathcal{J}(\theta)$ adjusts the policy π_θ to place more probability on higher–reward candidates, matching the inference-time selection rule. Since the likelihood ratios and KL term are computed within the same discrete candidate space, the optimization directly improves the distribution used for inference.

4) Reward Design: To achieve reliable single-shot selection under strong intra-class interference, we use a reward function for each candidate. As illustrated in Fig. 4, the scorer combines four items, including (L, S, D, E), each of which is mapped to $[0, 1]$ and then linearly aggregated with weights that sum to one, forming the total reward r_i as follows:

$$r_i = \sum_{k \in \{L,S,D,E\}} \lambda_k\, r_i^k, \tag{7}$$

where λ_k denotes the aggregation weight for component k.

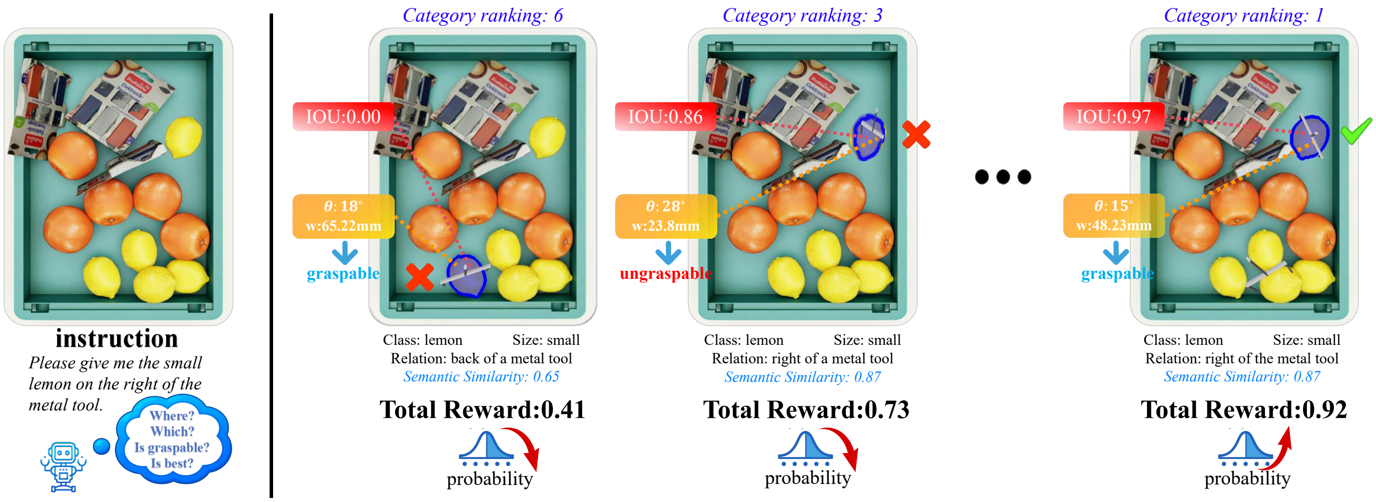

Fig. 4. **L/S/D/E Reward Mechanism in Training Stage.** For each candidate, we compute the Spatial Localization reward r^L from the mask IoU, the Semantic Alignment reward r^S from text–image similarity, the Intra-Class Disambiguation reward r^D from the margin to intra-class competitors, and the Planner-Free Executability reward r^E from the pose width w and approach angle θ. The total reward r is obtained by weighted summation of the four rewards, converting r to advantages and optimizing DRPO so that the policy's categorical distribution assigns higher probability to candidates with larger rewards.

The rewards within a scene are finally standardized across the group to produce advantages, which are consumed by the DRPO objective.

a) Spatial Localization Consistency (L).: Spatial localization consistency measures how well a candidate region matches the ground-truth target area indicated by the instruction. Denote $M_{\text{ref}}(q)$ as the ground-truth target mask, the spatial localization reward r_i^L is defined as:

$$r_i^L = \operatorname{IoU}\big(M_{\text{ref}}(q),\, m_i\big), \tag{8}$$

where IoU is the intersection-over-union metric.

This term translates the linguistic pointer in g into a concrete spatial constraint and favors candidates whose support coincides with the instruction-indicated area. In instruction-conditioned inference grasping, the grasp pose p_i is synthesized on top of m_i; therefore, aligning m_i to $M_{\text{ref}}(q)$ is critical to avoid selecting a intra-class but wrong instance, and it stabilizes ranking among near-duplicate objects. The signal remains effective in cluttered scenes because it evaluates relative overlap in the scene rather than raw appearance.

b) Semantic Alignment to the Instruction (S).: This reward asks whether the candidate region truly answers the object described by the instruction at the instance level [39]. We map the masked RGB patch and the text into a shared embedding space and score their agreement, ultimately obtaining the semantic alignment reward r_i^S. To that end, we first extract a visual embedding from the masked region and a textual embedding from the instruction:

$$\mathbf{v}_i = \phi_{\text{img}}(O \odot m_i), \qquad \mathbf{t}_g = \phi_{\text{text}}(g), \tag{9}$$

Here, ϕ_{img} and ϕ_{text} denote the image and text encoders that project inputs into the shared embedding space, $\odot$ denotes element-wise multiplication.

Masking binds the representation to the candidate itself and filters out distractors, which is essential when several intra-class instances appear in the scene. Next, we apply ℓ_2 normalization so that cosine similarity reduces to a dot product.

$$\widehat{\mathbf{v}}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|_2}, \qquad \widehat{\mathbf{t}}_g = \frac{\mathbf{t}_g}{\|\mathbf{t}_g\|_2}. \tag{10}$$

Finally, the cosine score is converted to a Semantic Alignment reward r_i^S by an affine map using the following equation:

$$r_i^S = \frac{1}{2}\Big(\widehat{\mathbf{v}}_i^{\top}\widehat{\mathbf{t}}_g + 1\Big) \in [0,1]. \tag{11}$$

In instruction-conditioned inference grasping, many failures arise from selecting an intra-class instance that violates the attributes specified by the instruction.

The alignment score compels the policy to honor the linguistic attributes encoded in the instruction, which separates visually similar instances and supplies a ranking signal that cannot be obtained from geometry alone. By enforcing what the region is, the reward reduces false picks and follow-up corrections, thus improving efficiency exactly when intra-class interference is strong.

c) Intra-Class Disambiguation (D).: In scenes with strong intra-class interference, several candidates may already satisfy spatial localization and semantic alignment, yet only one should be executed. We impose a within-class ordering that converts small differences easier to compare. First, location and appearance are fused into a single match score.

$$s_i = (1-\alpha)\, r_i^L + \alpha\, r_i^S, \quad \alpha \in [0,1]. \tag{12}$$

Subsequently, we define the intra-class competitor $\mathcal{C}_i$ set by a semantic threshold τ_S and exclude the candidate itself.

$$\mathcal{C}_i = \left\{ j \mid 1 \le j \le N,\ j \ne i,\ r_j^S \ge \tau_S \right\}, \quad \tau_S \in (0,1). \tag{13}$$

The strongest competitor score is then computed as b_i:

$$b_i = \begin{cases} \max_{j \in \mathcal{C}_i} s_j, & \mathcal{C}_i \ne \varnothing, \\ s_i, & \mathcal{C}_i = \varnothing, \end{cases} \tag{14}$$

from which the margin can be converted into an Intra-Class Disambiguation reward r_i^D.

$$r_i^D = \sigma\Big(\frac{s_i - b_i}{\tau_D}\Big), \\ \sigma(x) = \frac{1}{1+e^{-x}}, \tag{15}$$

where $\tau_D > 0$ is a temperature scaling the margin $(s_i - b_i)$.

This design focuses comparison on genuinely relevant competitors and rewards a positive lead, yielding a clear preference when several near-duplicates look equally plausible. Ties collapse toward neutrality and deficits are suppressed, which reduces mis-picks and repeated attempts. This reward complements localization and semantic alignment by helping select the correct instance when multiple candidates have similar location and semantic scores. After group-wise standardization, it integrates into DRPO to restore single-shot efficiency, even under strong intra-class interference.

d) Planner-Free Grasp Executability (E).: This component gives low scores to candidates that are unlikely to be executed successfully without invoking a motion planner, which helps the policy rank feasible candidates higher. Given a candidate pose $p_i = (\mathbf{t}_i, \mathbf{R}_i, w_i)$, we compute two soft gates [40]. The first gate softly measures whether the opening lies within the usable range of the gripper, and the second gate gives lower scores to poses where the gripper approaches the object from a highly tilted or sideward direction. We use the gripper's local z-axis $\hat{\mathbf{z}}$ as its approach axis. For each candidate pose p_i, the actual approach direction in the world frame is $\mathbf{R}_i\hat{\mathbf{z}}$. We compare it with the world-frame gravity direction $\hat{\mathbf{g}}$. The soft gates are defined as:

$$\widetilde{G}_w = \sigma\left(\frac{w_i - w_{\min}}{\tau_w}\right)\sigma\left(\frac{w_{\max} - w_i}{\tau_w}\right), \\ \widetilde{G}_a = \sigma\left(\frac{(\mathbf{R}_i\hat{\mathbf{z}})^\top\hat{\mathbf{g}} - \cos\alpha_{\max}}{\tau_a}\right), \tag{16}$$

where $\sigma(\cdot)$ denotes the sigmoid function. Here, $w_{\min}$ and $w_{\max}$ are the minimum and maximum usable opening widths of the gripper, and $\alpha_{\max}$ is the maximum allowed angle between the gripper approach direction and the gravity direction. The parameters τ_w and τ_a control the smoothness of the two soft gates. The planner-free executability reward r_i^E is then defined as a continuous feasibility score:

$$r_i^E = \widetilde{G}_w \cdot \widetilde{G}_a \in (0,1). \tag{17}$$

IV. Experiments and Analysis

A. Setup

FreeGraspData [15] is a static benchmark for language-conditioned grasping in cluttered tabletop scenes. It comprises 900 scenarios, and includes intra-class multi-instance scenes that pose challenges for referential understanding [41] and obstruction. Experimentally, FreeGraspData can be partitioned into two **splits** on aspects of *official* ambiguity: the ***unambiguous*** **split** (w/o Amb) admits a single clear referent for the instance, whereas the ***ambiguous*** **split** (w Amb.) contains multiple visually similar instances that plausibly match the description, in which identifying the correct target may need fine-grained language cues. This design makes FreeGraspData particularly suitable for validating methods that explicitly model discriminative language grounding and intra-class disambiguation in cluttered scenes. Each target includes three free-form instructions with varied wording and spatial references. Our model is finetuned on GraspNet data [2], and comparative and ablation experiments are conducted on FreeGraspData. The input for each scenario consists of a top-down RGB image and a natural language instruction.

TABLE I
EXPERIMENTS ON FREEGRASPDATA. HIGHER IS BETTER FOR SSR/RSR. NUMBERS ARE MEAN±STD AVERAGED. THE BEST RESULTS OF EACH COLUMN ARE HIGHLIGHTED IN **BOLD**.

Method	Reas.	Segm.	Metric	w/o Amb.	w Amb.
ThinkGrasp [17]	✓	✓	SSR	0.27 ± 0.02	0.26 ± 0.02
FreeGrasp [15]	✓	✓	SSR	0.39 ± 0.03	0.37 ± 0.03
LISA [19]	✓	✓	SSR	0.53 ± 0.01	0.47 ± 0.03
VisionReasoner [42]	✓	✓	SSR	0.62 ± 0.01	0.53 ± 0.02
RoboGround [43]	✓	✓	SSR	0.60 ± 0.04	0.56 ± 0.03
Ours	✓	✓	SSR	$\mathbf{0.66 \pm 0.03}\uparrow$	$\mathbf{0.61 \pm 0.02}\uparrow$
FreeGrasp [15]	✓		RSR	0.50 ± 0.02	0.41 ± 0.04
LISA [19]	✓		RSR	0.56 ± 0.01	0.52 ± 0.01
VisionReasoner [42]	✓		RSR	0.65 ± 0.03	0.56 ± 0.02
RoboGround [43]	✓		RSR	0.66 ± 0.04	0.57 ± 0.04
Ours	✓		RSR	$\mathbf{0.75 \pm 0.02}\uparrow$	$\mathbf{0.67 \pm 0.04}\uparrow$

TABLE II
ABLATION OF REWARD COMPONENTS OVER TWO SPLITS (WITH/WITHOUT AMBIGUITY). COMPARISON IS PERFORMED AMONG SFT, FOUR SINGLE-TERM ABLATIONS (W/O r^L, r^S, r^D, r^E), AND FULL MODEL.

Variant	RSR		SSR	
	w/o Amb.	w Amb.	w/o Amb.	w Amb.
SFT	0.48	0.42	0.67	0.47
w/o r^L	0.58	0.54	0.51	0.48
w/o r^S	0.64	0.61	0.58	0.53
w/o r^D	0.68	0.59	0.61	0.56
w/o r^E	0.73	0.65	0.61	0.57
Ours	**0.75**$\uparrow$	**0.67**$\uparrow$	**0.66**$\uparrow$	**0.61**$\uparrow$

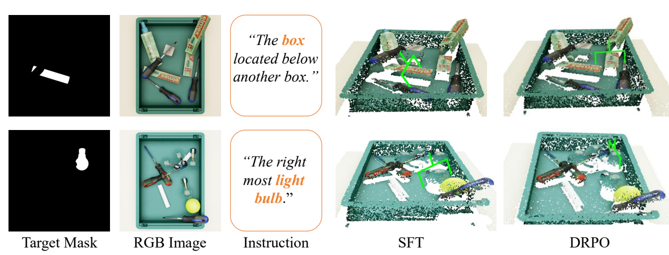

Fig. 5. Qualitative comparison of SFT and DRPO on ambiguous scenes.

B. Results on FreeGraspData

Following prior work [15], we evaluate our method using two static metrics. The **Reasoning Success Rate (RSR)** quantifies reasoning accuracy: the result is considered correct when the predicted target ID is present within the acceptable ground-truth set of IDs. The **Segmentation Success Rate (SSR)** assesses segmentation quality, requiring the IoU between the predicted mask of the selected object and its corresponding

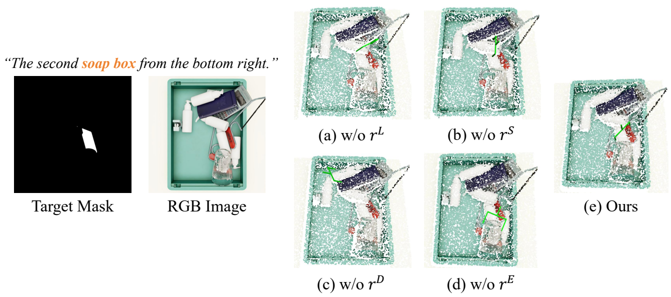

Fig. 6. Qualitative results of reward ablation.

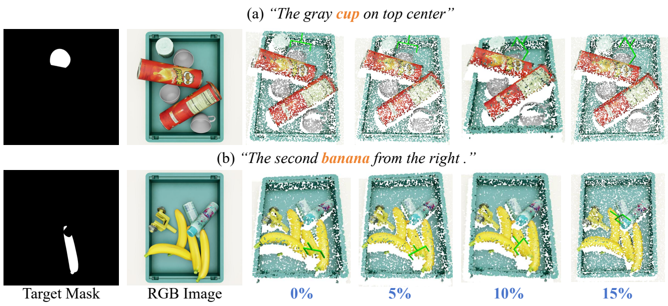

Fig. 7. Qualitative results under different instruction noise ratios.

ground-truth mask to exceed 0.5; this is computed independently of the reasoning outcome. For both metrics, we first compute the mean and standard deviation (mean $\pm$ std) across three instructions per object, and then aggregate these results over the official dataset splits. We use the model pretrained on GraspNet [2] dataset and directly evaluate its performance on FreeGraspData benchmark.

1) Comparisons with Other Methods: As shown in Table I, our method delivers the best overall SSR and RSR on FreeGraspData across the official splits under both ambiguity settings. Our key advantage is that the policy is trained to produce a discriminative ranking over region–pose candidates using group-normalized rewards, which explicitly sharpens preference between the correct referent and hard intra-class negatives guided by language. On the *unambiguous* split, we obtain gains of **4.0%** in SSR and **9.0%** in RSR, demonstrating reliable language grounding when a clear referent exists. In *ambiguous* scenes containing multiple intra-class instances, our performance improves further Compared with the strongest prior method RoboGround, we achieve **5.0%** higher SSR and **10.0%** higher RSR on the ambiguous split, reflecting more reliable intra-class disambiguation.

2) Ablation Studies: Under a setting matched to the supervised-only baseline, we train the policy with DRPO. Unlike supervised learning, which provides no explicit in-scene ranking and may bias the model toward a single annotated target, DRPO introduces task-verifiable rewards to encourage discriminative selection among language-grounded candidates. As shown in Table II, this procedure delivers consistent improvements over supervised-only training. Fig. 5 demonstrates the qualitative comparisons on ambiguous scenes.

To further disentangle the contribution of each reward term, we ablate the composite reward by removing one component at a time. As shown in Table II, eliminating any component consistently reduces both SSR and RSR, confirming that these four components are complementary. Removing r^L disrupts spatial localization, causing masks to drift from the instruction-indicated region, as shown in Fig. 6 (a). Removing r^S weakens language-vision alignment, yielding class-correct but attribute-inconsistent selections, as shown in Fig. 6 (b). Removing r^D narrows the reward gap among intra-class candidates; when the reward values of different candidates are nearly tied, the group-normalized advantages become small and DRPO conveys little information to improve the ranking, as shown in Fig. 6 (c). Removing r^E eliminates feasibility gating, so rewards no longer penalize infeasible poses, thereby increasing the incidence of unstable grasp poses, as shown in Fig. 6 (d). In contrast, the full reward leads to a more accurate target selection and a more stable grasp pose, as shown in Fig. 6 (e).

3) Robustness to Noisy and Ambiguous Instructions: We conduct a controlled study to evaluate the robustness of the policy to ambiguity in language instructions. Starting from the split with unambiguous instructions, we randomly select a subset of scene–instruction pairs and replace their instructions with more ambiguous variants. For example, an instruction such as "the second lemon from the left" can be changed to "the lemon", so that several lemons in the scene may match the instruction. The underlying scenes, instance segmentations, and target labels are kept fixed. The noise ratio specifies the fraction of instructions that are replaced. We consider noise ratios of 0%, 5%, 10%, and 15%. For the NOISY variant in our ablations, such perturbations are injected only into the training instructions, while all evaluation instructions remain clean and unchanged.

For each noise level, we keep the policy fixed, re-evaluate it, and report RSR and SSR on the ***unambiguous*** and ***ambiguous*** splits, as shown in Fig. 9. As the noise ratio increases from 0% to 15%, both metrics on the ***unambiguous*** split drop by approximately four percentage points, while the ***ambiguous*** split exhibits a similar monotonic decline. The unambiguous split remains better than the ambiguous split at all noise levels. Qualitative examples under different noise ratios are shown in Fig. 7. In Fig. 7(a), the model consistently selects the correct target under all noise ratios. In Fig. 7(b), the model still selects the intended banana at 0%, 5%, and 10% noise, with an incorrect selection appearing only at 15% noise. These results show that the policy remains stable under moderate instruction noise, and that such noise does not lead to an obvious performance drop.

These observations indicate that the policy maintains a reasonable level of robustness under moderate instruction corruption. DRPO optimizes a distribution over discrete region–pose candidates and relies on a discriminative reward that combines localization, semantic alignment, disambiguation, and executability. When relational cues in the text become partially unreliable, the localization and alignment terms carry weaker signals, but the disambiguation and executability components still promote candidates that correspond to physically feasible grasps on plausible targets. As a result, the policy continues to prefer consistent region–pose pairs despite an increasing proportion of ambiguous instructions, and the degradation in RSR and SSR remains gradual.

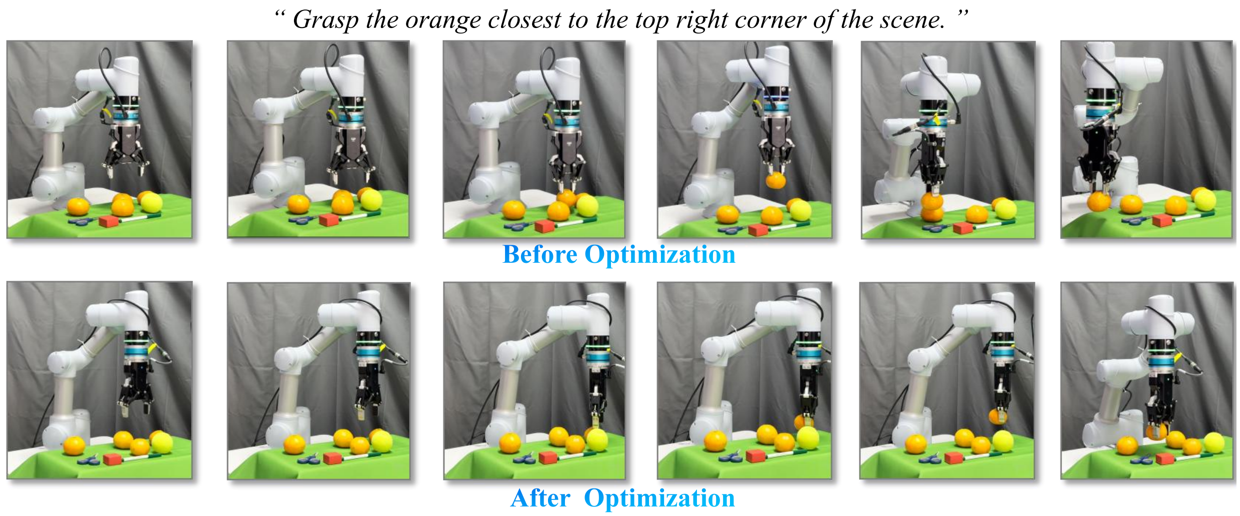

Fig. 8. Grasping results before (top) and after (bottom) policy optimization under intra-class ambiguity.

4) Inference Time Analysis: We evaluate end-to-end inference latency on the same workstation equipped with an Intel Xeon w7-3455 CPU and a single NVIDIA RTX A6000 GPU. All methods use the same input resolution and RGB-D preprocessing protocol. Latency is measured from receiving an RGB-D scene and a language instruction to outputting a single 6-DoF grasp pose, averaged over 100 episodes. As shown in Fig. 10, FreeGrasp requires 16.40 s per input sample, followed by ThinkGrasp with 12.03 s. LISA and VisionReasoner require 8.91 s and 9.14 s, respectively, whereas our method achieves the lowest latency of 7.88 s. These results show that the proposed DRPO-based pipeline reduces inference time while remaining practical for deployment on a single machine.

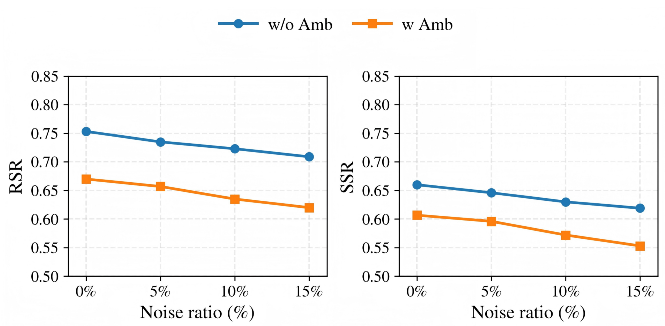

Fig. 9. **Robustness to noisy instructions.** RSR and SSR under increasing instruction noise ratios.

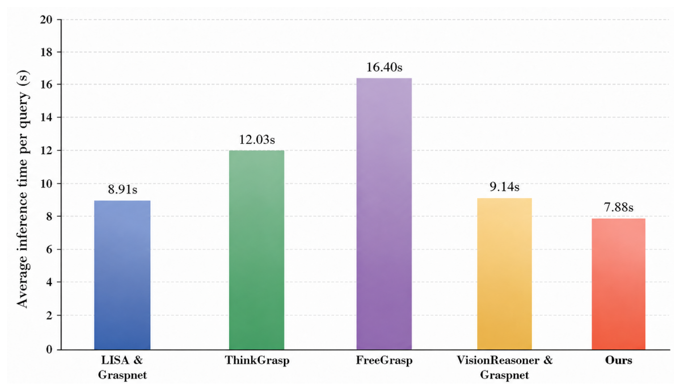

Fig. 10. **Inference time comparison.** Average end-to-end inference time per query for different language-driven grasping pipelines, where lower values indicate faster inference.

C. Real Robot Applications

We evaluate our language-driven grasping in a physical setup consisting of a SIASUN manipulator, DAHUAN AG95 gripper, and an Intel RealSense D435i camera.

As exemplified in Fig. 11, intra-class interference is defined by the number of distractors sharing the target category, denoted as K_{dist}. Scene difficulty is classified into **Simple** (K_{dist}=0; no intra-class distractors), **Basic** ($K_{\text{dist}}\in\{1,2\}$), and **Advanced** ($K_{\text{dist}}\geq 3$). Each instruction specifies basic attributes, which is human-authored and is verified to ensure a unique ground-truth target. The evaluation covers 30 scenes per difficulty level, reporting the overall success rate (SR) as the comparison metric.

As shown in Table III, compared with other methods, our policy exhibits superior performance. In straightforward scenes, performance is broadly comparable across methods; however, once multiple instances of the target class are present, the gap becomes evident. In scenes with multiple intra-class instances, the compared methods often select an object that matches the class but violates the specified attribute constraints, resulting in mis-selections.

Fig. 8 contrasts grasp behaviors before and after policy optimization on two hard real-robot scenes with strong intra-class interference. In this instruction, "the top right corner" refers to the upper-right region of the RGB image as viewed by the camera, rather than the robot's physical right side. Table IV compares the single-shot success rates (SR) before and after optimization across difficulty levels. Before optimization, the policy often focuses on same-category distractors and incorrectly selects near-duplicate objects that do not match the instruction, even when the resulting grasp poses are physically plausible. After optimization, it more reliably identifies the intended target in the presence of intra-class distractors and markedly reduces such mis-selections, leading to more reliable

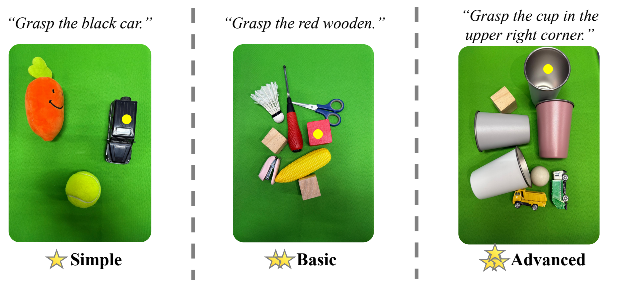

Fig. 11. Examples of Simple, Basic and Advanced scenes. The yellow dots indicate the target objects to be grasped.

single-shot execution.

To further assess the role of each reward component in physical execution, we conduct a real-robot ablation study by removing one reward term at a time. As reported in Table V, each component contributes to the final single-shot success rate. The performance differences become more pronounced in the Basic and Advanced settings, where the presence of multiple intra-class distractors makes target selection more challenging. In particular, removing the intra-class disambiguation reward results in a noticeable performance drop under strong interference, indicating its importance for distinguishing the intended target from similar objects. Removing the executability reward also degrades real-robot performance, suggesting that pose feasibility is critical for reliable physical grasping. These results show that the proposed reward design improves not only offline reasoning accuracy but also the practical reliability of language-driven robotic grasping.

TABLE III
SINGLE-SHOT SUCCESS RATE (SR) ON REAL-ROBOT INSTRUCTIONS ACROSS DIFFICULTY LEVELS. BEST VALUE IN EACH COLUMN IS IN **BOLD**.

Method	Simple	Basic	Advanced
LISA [19] & GraspNet [2]	0.53	0.43	0.13
ThinkGrasp [17]	0.70	0.43	0.23
FreeGrasp [15]	0.73	0.53	0.20
VisionReasoner [42] & GraspNet [2]	0.73	0.57	0.27
Ours	**0.80**↑	**0.67**↑	**0.47**↑

TABLE IV
SINGLE-SHOT SUCCESS RATE (SR) ON REAL-ROBOT INSTRUCTIONS ACROSS DIFFICULTY LEVELS BEFORE AND AFTER POLICY OPTIMIZATION. BEST VALUE IN EACH COLUMN IS IN **BOLD**.

Method	Simple	Basic	Advanced
Before Optimization	0.60	0.37	0.23
After Optimization	**0.80**↑	**0.67**↑	**0.47**↑

V. LIMITATIONS AND FUTURE WORKS

While DRPO significantly improves success rates under strong intra-class interference and performs well in real-robot grasping, several limitations remain. In highly cluttered scenes, severe occlusions often produce fragmented or missing masks. Consequently, the correct region–pose pairing may be absent from the candidate set, imposing an inherent upper bound on any ranking-based selection. Occlusion also yields incomplete geometry, especially under tight clearances, which can bias feasibility checks and generate grasps that appear valid at inference time but collide during execution.

Future work will explore occlusion-aware instance refinement and relation-centric grounding for more reliable disambiguation, and will incorporate lightweight clearance and collision constraints with multi-view sensing to reduce uncertainty and improve robustness in dense clutter.

TABLE V
SINGLE-SHOT SUCCESS RATE (SR) OF REAL-ROBOT REWARD ABLATIONS ACROSS DIFFICULTY LEVELS. THE BEST VALUE IN EACH COLUMN IS HIGHLIGHTED IN BOLD.

Method	Simple	Basic	Advanced
w/o r^L	0.67	0.53	0.33
w/o r^S	0.73	0.57	0.37
w/o r^D	0.77	0.53	0.30
w/o r^E	0.73	0.63	0.40
Ours	**0.80**↑	**0.67**↑	**0.47**↑

VI. CONCLUSION

In this work, we reformulate language-driven robotic grasping as a top-k selection task and leverage reinforcement learning with rule-defined and verifiable rewards. This approach is instantiated in our proposed Discriminative Reward Policy Optimization (DRPO), which introduces four carefully designed rewards, i.e., spatial localization, semantic alignment, intra-class disambiguation, and grasp executability, to directly align policy optimization with desired grasps. By learning from ranking-based, non-differentiable, and task-verifiable feedback that conventional supervised fine-tuning cannot capture, DRPO effectively enhances relational reasoning in complex multi-object scenarios. Extensive evaluations on both relational-reasoning benchmarks and real-world robotic setups demonstrate the effectiveness of our approach: it achieves state-of-the-art performance in Reasoning Success Rate on the FreeGraspData and consistently grasps the target object among multiple same-category instances in real-robot experiments, validating its practical reliability and generalization.

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